Technical Information
Fuel & Lubricant Solutions

BASF
We create chemistry

OPPANOL® C grades

OPPANOL® C grades are intended for use in producing adhesives, sealants, lubricating oils, and coating compounds for food packaging. They are a chipped version of OPPANOL® N grades.

*For more information of OPPANOL® N grade properties, please refer to TI/EVO 3220.

® = registered trademark of BASF SE

Chemical composition high molecular weight polyisobutenes with a talc-based separating agent

Properties

Product range

OPPANOL® C	80	100	150
Separating agent [%] (average weight percentage)	8.0	5.0	5.0
Stabilizer [ppm] (average BHT concentration)	500	500	500
Specification upon delivery			
Staudinger index J_0 [cm³/g]	178 – 236	241 – 294	416 – 479

The Staudinger Index J_0 [cm³/g] is calculated from the flow time at 20 °C for the solution through the Capillary of an Ubbelohde viscometer.

$$J_0 = \frac{\eta\, sp}{c(1 + 0,31 * \eta\, sp)}$$

(Schulz-Blaschke-equation)

$$\eta\, sp = \frac{t}{t_0} - 1$$

(specific viscosity)

$t = running\ time\ of\ Oppanol\ solution;$ applying Hagenbach-Couette correction
$t_0 = running\ time\ of\ isooctane,$ applying Hagenbach-Couette correction
$c = Concentration\ of\ Oppanol\ solution\ in\ g/cm^3$

Typical characteristics

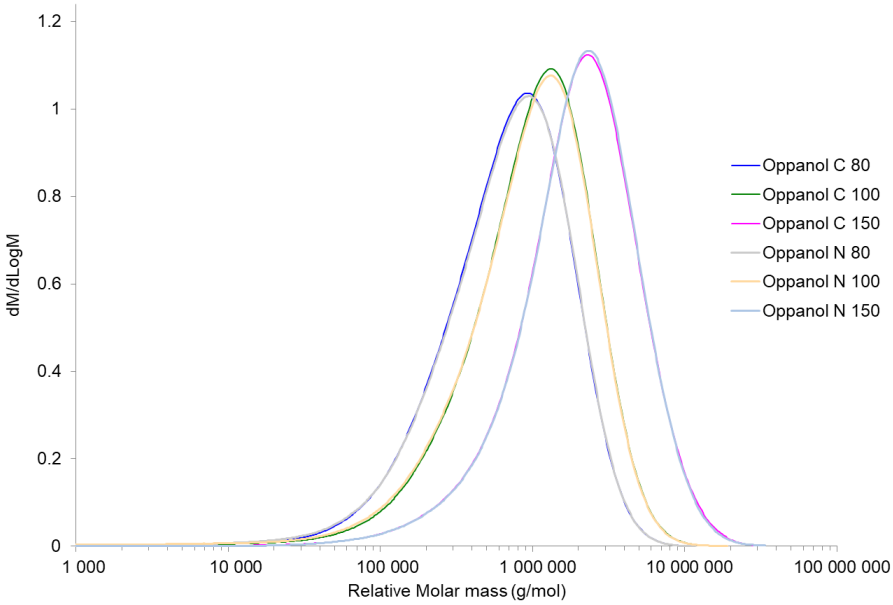
OPPANOL® C	80	100	150
Average molecular weight M_V (viscosity average)	800,000	1,110,000	2,600,000
Chip size [inch]	< 1.2		
Color	White to tan or light grey		

The viscosity average molecular weight is calculated according to the formula

$$M_V = \sqrt[0,65]{\frac{J_0 * 10^2}{3,06}}$$

Comparison to OPPANOL® N

OPPANOL® C grades are not expected to undergo molecular weight degradation during the chipping process.



Example: Gel permeation chromatograph of OPPANOL® C and OPPANOL® N grades

Solubility

The OPPANOL® C grades may be dissolved in aliphatic, aromatic, cyclic, and halogenated hydrocarbons (the separating agent will be insoluble and can be removed through filtration). Their tendency to swell on solvation with alcohols, ethers, esters, and ketones increases with the length of the hydrocarbon chain. The viscosity of solutions of OPPANOL® C grades differ according to the solvent in question and its solvation behavior.

See “Safety” section for handling and disposing of talc (not containing asbestos) in powder form.

Composition

OPPANOL® C grades consist primarily of polyisobutene except for stabilizer and a separating agent.

Stabilizer and separating agent

The OPPANOL® C portfolio includes stabilized grades with a minimally incorporated separating agent. In order to prevent the polymer from oxidation, stabilized grades contain an average of approximately 500 ppm 2,6-di-tert-butyl-4-methylphenol (BHT). For outdoor applications, the addition of UV stabilizers like UV absorbers or steric hindered amines is recommended.

To inhibit cold flow, OPPANOL® C grades contain an average of approximately 5 – 8 weight % of a talc-based separating agent.

Quality control

The above-listed data represent average values at the time of going to press of this Technical Information. They are intended as a guide to facilitate handling and cannot be regarded as specified data. Specified product data are issued as a separate product specification.

Processing

Blending

There are two ways to blend OPPANOL®: Dissolving and Melting.

Dissolving: OPPANOL® C grades are soluble in aliphatic, aromatic, cyclic and halogenated hydrocarbons (the separating agent will be insoluble and can be removed through filtration). They are swollen by alcohols, ethers, esters and ketones. The extent of swelling is increasing with the length of the hydrocarbon chain.

See “Safety” section for handling and disposing of talc (not containing asbestos) in powder form.

Melting: OPPANOL® C grades are thermoplastics and can be processed with conventional machinery used in the rubber industry, including kneaders, roll-mills and single- or twin-screw extruders. Sigma-bladed mixers are widely used and, depending on the viscosity range, even Banbury mixers.

Formulation

OPPANOL® C grades are mostly used in formulations into which they provide their properties.

Formulations are stepwisely prepared melt- or solvent-based mixtures that include one or more OPPANOL® grades. They may contain fillers, pigments, stabilizers and other polymers (e.g. butyl rubber, hydrocarbon resins and others) that are compatible with polyisobutene.

Equipment

Due to its size, OPPANOL® C can be fed into common manufacturing equipment without additional processing.

For the processing of the OPPANOL® C grades, the following equipment can be used: **Kneaders** which can vary in the chamber size (often adjusted to PIB weight), rotor types, torque, speed, etc.; **Extruders** which can vary the screw types, the alloy, with or without vacuum, etc.; **Mixers**; **Calenders** (for final compounds)

Remark: Starting material should be the high viscosity component with stepwise addition of the lower viscosity component. Mixing time should be adjusted accordingly.

Safety

General

When using this product, the information and advice given in our Safety Data Sheet should be observed. Due attention should also be given to the precautions necessary for handling chemicals.

In particular, the place of work must be well ventilated if large quantities are being processed, the skin should be protected, respiratory protection and safety glasses should be worn at all times.

Talc (not containing asbestos) is not considered a hazardous waste as defined by the US EPA RCRA (40 CFR 261) regulations. Observe all applicable federal, state, and local regulations when handling, storing, or disposing of this substance.

Safety Data Sheet

Safety Data Sheet have been compiled for all OPPANOL® C grades that contain up-to-date information on all questions relevant to safety.

Labelling

According to all the data at our disposal, all OPPANOL® C grades are not dangerous substances or preparations as defined in the Globally Harmonized System of Classification and Labelling of Chemicals (GHS).

Industrial hygiene

According to our experience and information, OPPANOL® C grades do not exert any harmful effects on health, provided they are used for the purpose for which they are intended and processed in accordance with current industrial practice.

Refer to OSHA regulations for exposure limits when handling talc (not containing asbestos) in powder form. [CAS #14807-96-6]

Certificates

The OPPANOL® C grades are certified and manufactured according to the following quality standard: ISO 9001. Additional quality programs, industry standards and guidelines are applied to ensure product quality, safety, security and supply of the products according to various industry needs. Certificates and statements for OPPANOL® C grades regarding regional-, country- and industry-specific requirements are available upon request.

Packaging

50 lb. (22.7 kg) cardboard boxes containing a clear LDPE bag.

Shelf life

Storage at ambient temperatures and protection from light and moisture provided a shelf life (‘best before’ date) of six months from the date of production for all grades. Damage to packaging must be strictly avoided while in storage or during handling.

Note

The declaration and information given herein is exclusively provided for our customers and the respective competent authorities.

It is not intended for publication either in printed or electronic form (e.g. via Internet) by any third party. Neither partial nor full publication is allowed without the prior written permission of BASF.

The data indicated above are the results of our investigations, correspond to the state-of-the-art and are based on our current knowledge and experience. The data refer to the state of the laws at the date of issue.

BASF produces a wide variety of high quality polyisobutenes marketed by BASF under the trademark OPPANOL® that satisfy the manifold requirements of our customers, including products that may meet the specifications for use in food, medical, pharmaceutical or cosmetics applications.

BASF has proven expertise in supporting and working with our customers in the innovative use and application of our materials.

However, BASF has not designed or tested its OPPANOL® grades with respect to special requirements related to their use in medical devices (defined in the European, US or other local medical device legislation), pharmaceuticals and cosmetics.

In view of the many factors that may affect the processing and use of our OPPANOL®, the data in this publication do not relieve processors of the responsibility to carry out their own inspections and tests, neither do these data imply any guarantee of certain properties, nor the suitability of the product for a specific purpose.

Any descriptions, drawings, photographs, data, proportions, weights etc. given herein may change without prior notice and do not constitute the agreed contractual quality of the product. It is the responsibility of the recipient to ensure that all proprietary rights, laws and legislation are observed.

BASF does not recommend the use of or claim the suitability of OPPANOL® in a specific application and, therefore, the decision to use OPPANOL® is solely at the customer's own risk. It is the responsibility of the customer to determine whether their manufacturing process and the end application using OPPANOL® is safe, lawful and technically suitable for the intended use. BASF extends no warranties or guarantees, express or implied, concerning the suitability of OPPANOL® for any specific application, especially for a possible use in medical, pharmaceutical or cosmetics applications. Moreover, BASF does never supply its OPPANOL® products for the manufacture of implants.

February 2021

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